Enzymes

Catalyst: Substance that speeds up the rate of a chemical reaction and is not changed in the reaction.

Enzymes: Proteins that act as biological catalyst.

→ Enzymes speed up all metabolic reactions in the body which are divided into catabolic and anabolic reactions.

Catabolic reaction: A reaction that breaks substances down

$$C \longrightarrow A + B$$

Anabolic reaction: A reaction that builds substances up

$$A + B \longrightarrow C$$

→ The substance that the enzyme works on is substrate

→ The substance produced by a reaction catalysed by an enzyme is called a product.

* Protease breaks down protein to amino acids
* Amylase breaks down startch to glucose
* Lipase breaks down fat into three fatty acids and one glycerol
* Carbohydrase breaks down carbohydrates to glucose.

The lock and key model:

Substrate: key Enzyme: lock Active site: key hole

* Enzymes catalyse specific reactions

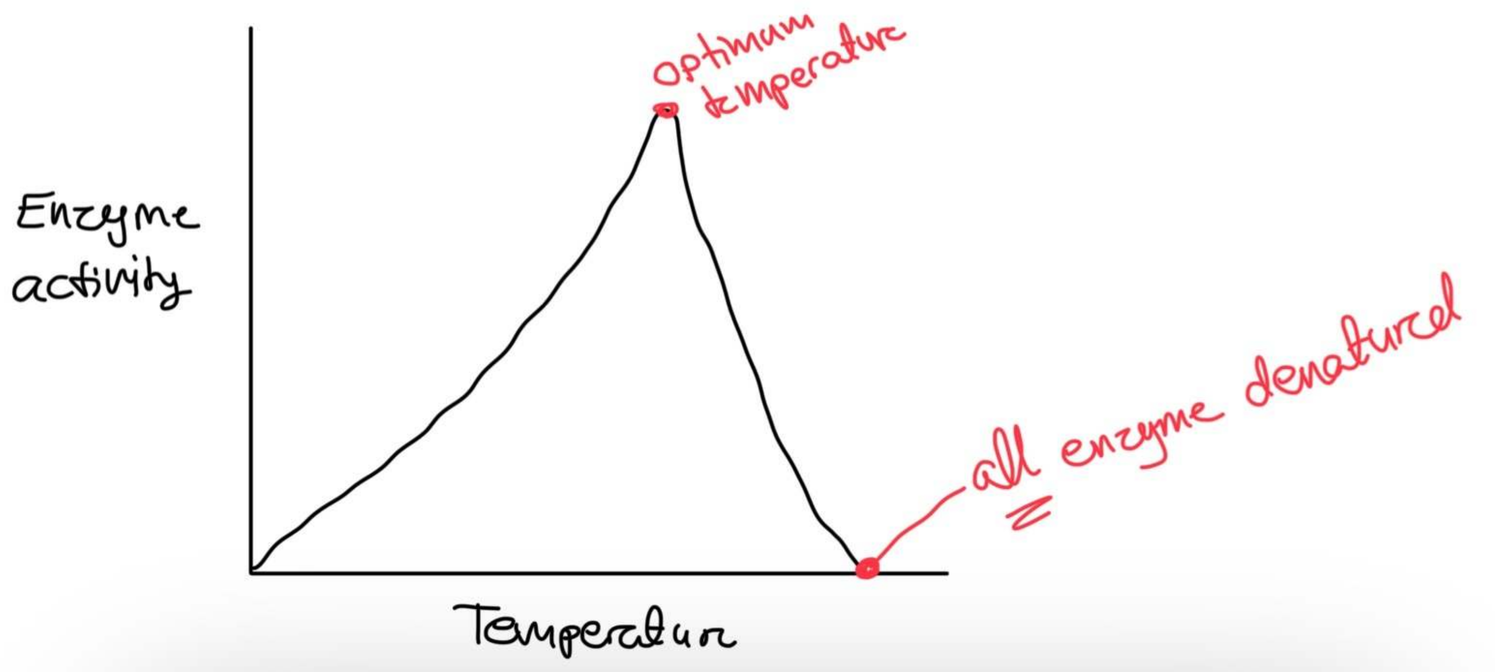
→ Enzymes are specific, they have specific shapes to their active sites. So only a particular substrate has a shape complementary to the active site.

→ The substrate fits into the active site forming enzyme-substrate complex. A product is formed and released from the active site.

* The enzyme can be reused again as it remains unchanged.

Effect of temperature on enzyme activity.

As the temperature increases, the rate of enzyme activity increases as molecules gain more kinetic energy and move faster. The substrate collides more often with the enzyme, resulting in more successful collisions. More substrates fit into the active site of the enzyme, forming more enzyme-substrate complex. Until the temperature reaches the optimum temperature, which is the temperature at which the enzyme is at its maximum activity. As the temperature increases above the optimum temperature, the enzyme becomes denatured: The shape of the active site changes. So the substrate can no longer fit the active site, and no enzyme-substrate complex forms, and so no products form.



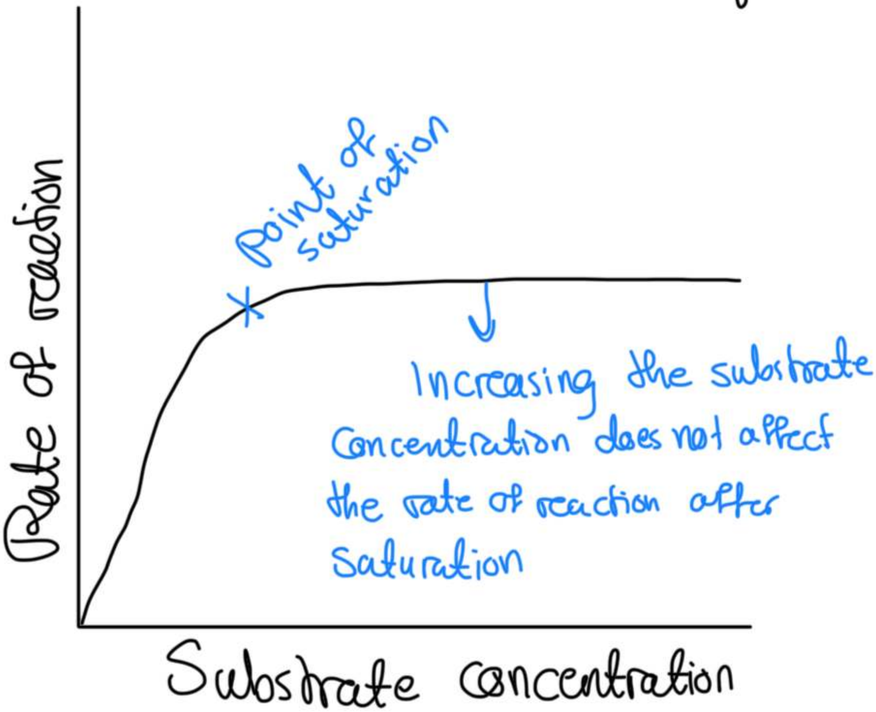
Effect of pH on enzyme activity

* Enzymes are at their maximum activity on their optimum pH. If the pH is no longer at the optimum level, higher or lower, the shape of the active site changes, and the enzyme becomes denatured. The substrate can longer fit into the active site of the enzyme, and no enzyme-substrate complex form → no products form

→ Most enzymes have a pH of 7

→ Pepsin, an enzyme that catalyses the break-down of proteins into polypeptides in the stomach, have an optimum pH of 2. This is because of the hydrochloric acid present in the stomach

The effect of substrate concentration on enzyme activity



As the concentration of substrate increases, the rate of enzyme activity increases, as substrates collide more with the enzyme's active site. So more enzyme-substrate complex forms. Up until a point where all the active sites of enzymes would be occupied by substrates, then increasing the substration concentration would not affect the enzyme activity $\longrightarrow$ remains constant